PURCHASE TOTAL WITH DISCOUNT AND TAX

AIM

To calculate a customer's total purchase cost after applying a discount and tax.

ALGORITHM

1. Start.
2. Enter item prices, quantities, discount rate, and tax rate.
3. Calculate subtotal by adding price × quantity for each item.
4. Subtract the discount from the subtotal.
5. Calculate tax on the discounted amount and add it to get the final total.
6. Display the final cost.
7. Stop.

PROGRAM (PYTHON)

```
prices = [100, 50, 200]
quantities = [2, 3, 1]
discount, tax = 10, 5  # percentages

subtotal = sum(p*q for p, q in zip(prices, quantities))
total = subtotal * (1 - discount/100) * (1 + tax/100)

print("Total cost:", round(total, 2))
```

INPUT

Item price	Quantity	Amount
100	2	200
50	3	150
200	1	200

Discount = 10%; Tax = 5%

OUTPUT

```
Total cost: 393.75
```

RESULT

The final purchase cost after a 10% discount and 5% tax is 393.75.